

hardware components that are readily available. Another cost-cutting factor is that the project runs on free and Open Source software, mostly written in Python programming language.

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Phoenix' hardware consists of an Atmel ATmega32 microcontroller running a 'C' program, a 12-bit resolution analogue-to-digital converter (ADC), a 12-bit digital-to-analogue converter, variable gain amplifiers, several oscillators and a programmable constant-current source. It is interfaced to a computer that runs Python.

"For the software, we have chosen Python, as it has a huge collection of library and programs for scientific computation. Several GUI programs have already been written for doing various experiments. A user can conduct several experiments using the GUI programs provided, but designing new experiments would require accessing the hardware accordingly. Hardware features can be accessed by entering codes that are well documented from the Python library," Dr Kumar explains.

Phoenix's clear-cut architecture is one of its USPs, enabling teachers to develop new experiments with the help of the library. Dr Kumar elaborates, "School teachers are usually not inclined to learn about electronics or micro-level C-programming, but this is usually required in communicating with the hardware. With Phoenix, all they need to do is

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write one line of code that is readily available in the Python Library, and they will get the required data. For example, if you want to capture the frequency from a guitar, you can type a code on the computer and send a request to the hardware unit. The computer's monitor will display the frequency while you are kept detached from the programming details."

The communication takes place between the microcontroller in the hardware unit and the Python-running computer through a USB connection.

Dr Kumar adds, "The earlier units that we designed had parallel ports but this kind of interfacing lost popularity. Hence we migrated to USB interface. It also helped that Python already had a library to communicate to the serial port."

Overcoming challenges

Ensuring a low price point of Phoenix was one of the biggest challenges before Dr Kumar and his team while designing the platform. Getting high-resolution components at low prices was a challenge in itself and required a lot of research. For example, IUAC team went to different markets to compare prices, ease of availability and resolution before coining down on ADCs. They decided to use a product from Microchip which was available for Rs 150.

"Apart from affordability, we looked for components that could be

sourced easily and repaired by the users in case of damages common in school and college laboratories," reasons Dr Kumar.

The IUAC team also worked on the flexibility of the framework by writing the software in a generic manner. It is easy to get the hardware and write a simple software that makes the platform rigid. However, this would mean limited number of experiments using the platform.

Dr Kumar illustrates, "We can easily design platforms that work for five experiments, but for the sixth one the user should know all the details of hardware design and coding. This makes the architecture rigid. In Phoenix, a lot of time and effort went into making the architecture flexible. Its generic nature allows software that supports time measurement, to be used in measuring acceleration due to gravity and the velocity of sound."

What lies ahead

IUAC's goal is to help engineering students develop their own projects rather than purchase from professionals in the field. It has approached NCERT and CBSE to make it a part of every school's lab activities.

"As of now, the only explanation given for AC and DC is that AC means alternating current and DC means direct current. Most of the teachers cannot answer a simple query like what is the nature of voltage on a three-point mains socket. We hope that Phoenix will help them rise by changing their practices, making students and teachers more aware of what is in their books," Dr Kumar concludes. ●

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The author is from EPF Bureau, Bengaluru